EchoSight-Pro

User Guidance

For version 0.5.1

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0. Preface

0.1 What is EchoSight-Pro

EchoSight-Pro is a medical ultrasound signal chain core framework, which is designed and developed by myself alone.
The design of EchoSight-Pro is quite engineering oriented and efficiency targeted.

0.2 Who can benefit from EchoSight-Pro

Theoretical Research People
For the Master and PhD students, as well as professors at universities or research institutions, you can use EchoSight-Pro to check and evaluate your research topics.

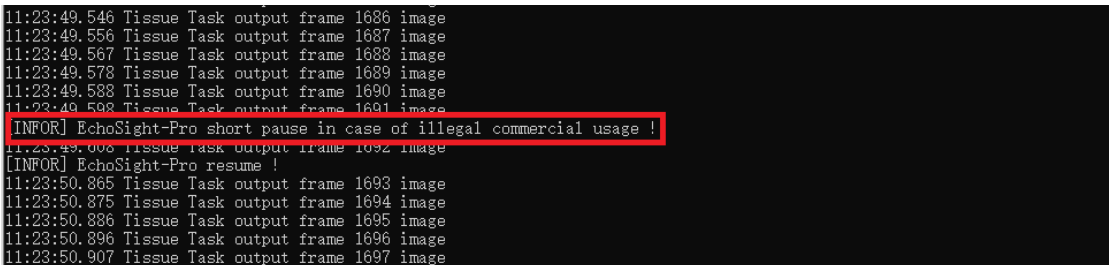
Ultrasound Start-up Corporation People
Creating an ultrasound machine from scratch is always of tons of uncertainties and big risks. You can use EchoSight-Pro to evaluate your hardware prototype by dumping your raw data before machine is built up. And it can give you confirmed answer to whether your industry computer is able to support software beam forming or not.
Also, EchoSight-Pro can be used to check if the channel data from your hardware board/FPGA is correct or not.

Traditional Ultrasound Corporation People
Two approaches of ultrasound commercial framework design exist at least. One is FPGA / hardware chip based; another is software based.
EchoSight-Pro can be used to testify the software RTB synthetic beam forming possibility, before your team starting to explore the new technique road path, and check your front end before you preparing project resource and recruit the team.

0.3 Copyrights

EchoSight-Pro is **free of charge** to use, as long as you do **NOT** try to integrate it into your commercial product. I keep all the copyrights of EchoSight-Pro.
If you have bugs to report or any specific needs from EchoSight-Pro, you can contact this email echosight_pro@126.com(Please feel no shame about that, coz I do not often check email neither.)
After running a period of time (about 19 seconds, more or less), EchoSight-Pro will pause for a short break (around 1 second). It is **NOT** software bug; I made it deliberately. This is designed to avoid illegal usage or integration for commercial purpose. I think won't affect

non-business usage.



1. Quick Start

1.1 Download and Run

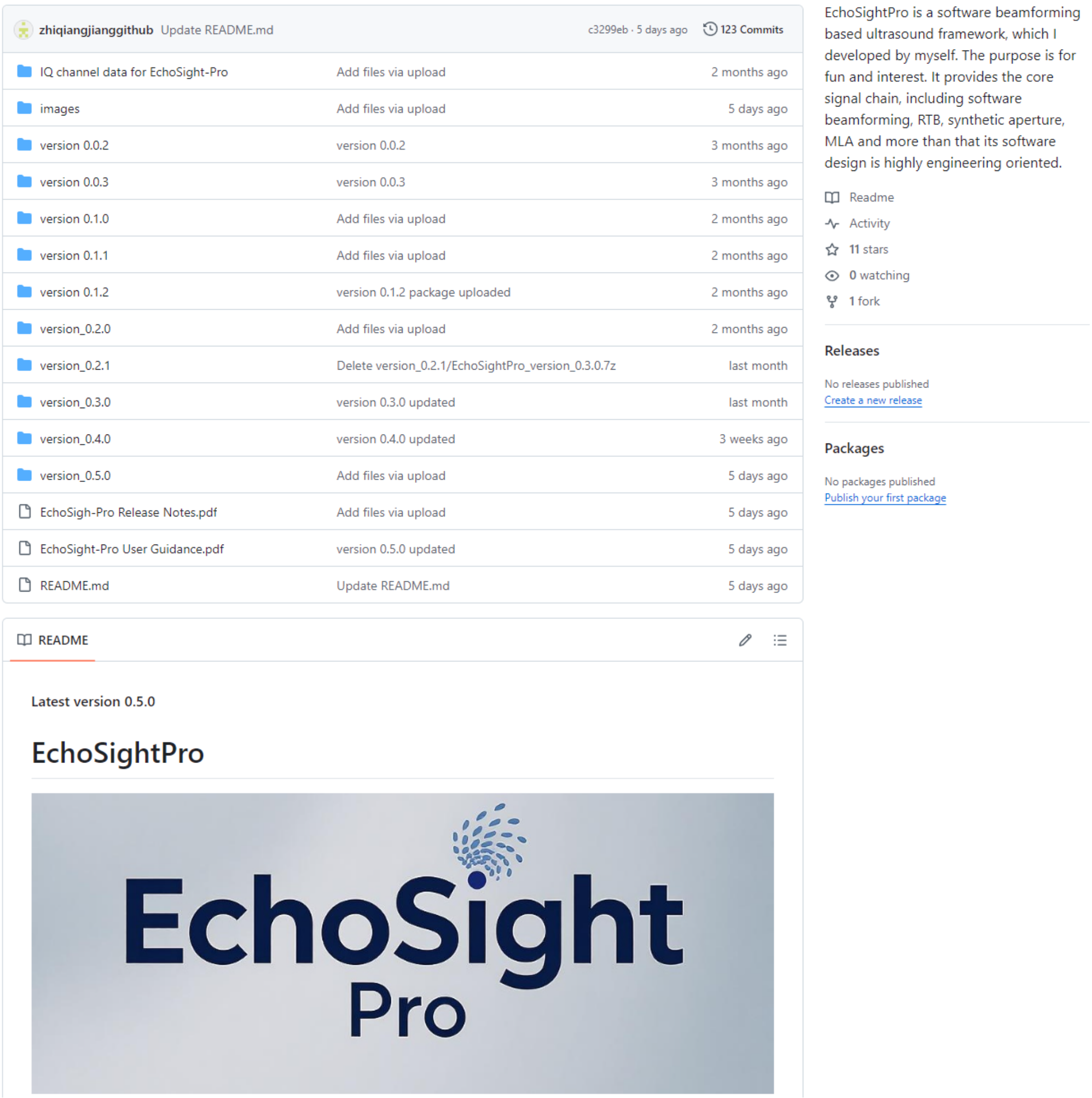
You can down load the software package from Github or CSDN. Both of them provide latest version download.

CSDN release page（简体中文）：

https://blog.csdn.net/gamer_gerald/article/details/149256620?spm=1001.2014.3001.5502

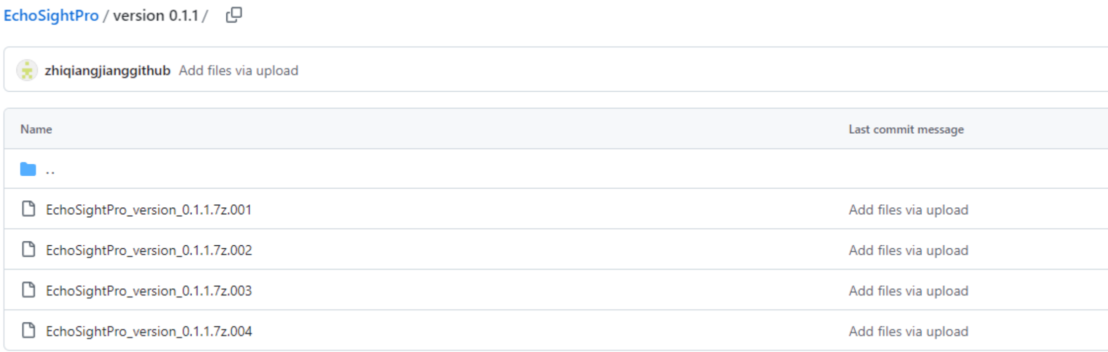
GitHub release page（English）：

https://github.com/zhiqiangjianggithub/EchoSightPro



On Github, all history versions are provided. Please download the latest version, current document might not match with old version software.

For versions before 0.2.x, you need to down load several zip files in the folder, before you unzip them (due to github 25MB single file limits).

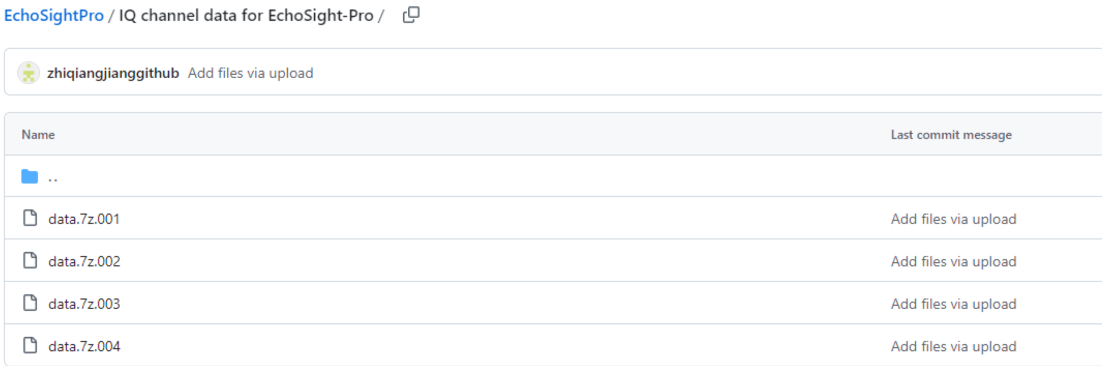


EchoSightPro / version 0.1.1 /

zhiqiangjianggithub Add files via upload

Name	Last commit message
..	
EchoSightPro_version_0.1.1.7z.001	Add files via upload
EchoSightPro_version_0.1.1.7z.002	Add files via upload
EchoSightPro_version_0.1.1.7z.003	Add files via upload
EchoSightPro_version_0.1.1.7z.004	Add files via upload

For version since 0.2.x or later, only 1 zip file for the software is needed. The channel data file is stored in another folder, so you can reuse them instead of download huge data every time. https://github.com/zhiqiangjianggithub/EchoSightPro/tree/main/IQ%20channel%20data%20for%20EchoSight-Pro



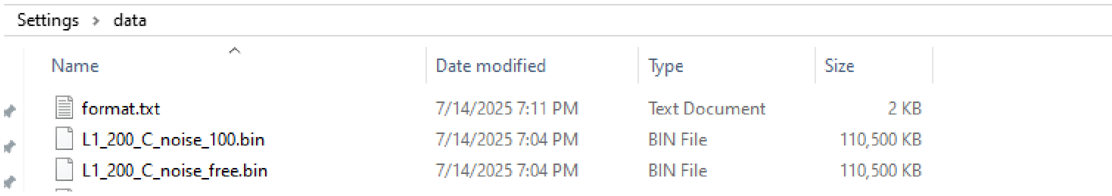
EchoSightPro / IQ channel data for EchoSight-Pro /

zhiqiangjianggithub Add files via upload

Name	Last commit message
..	
data.7z.001	Add files via upload
data.7z.002	Add files via upload
data.7z.003	Add files via upload
data.7z.004	Add files via upload

After download the package, please unzip it to the folder whatever you like. It's pure green software, no installation needed.

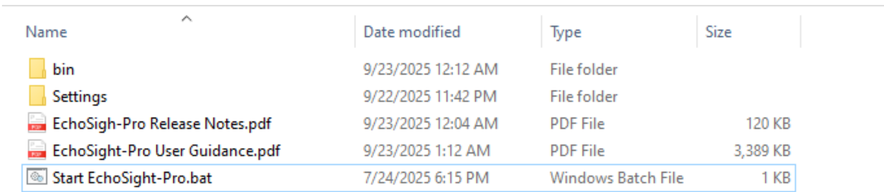
Make sure the channel data file is put into the right folder.



Settings > data

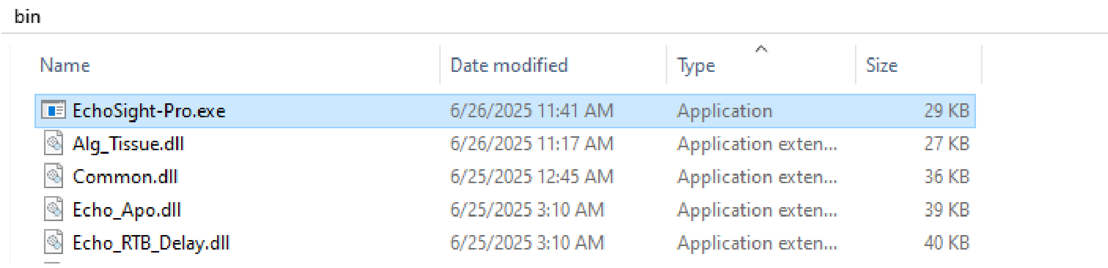
Name	Date modified	Type	Size
format.txt	7/14/2025 7:11 PM	Text Document	2 KB
L1_200_C_noise_100.bin	7/14/2025 7:04 PM	BIN File	110,500 KB
L1_200_C_noise_free.bin	7/14/2025 7:04 PM	BIN File	110,500 KB

Double click **Start EchoSight-Pro.bat** to start EchoSight-Pro program.



Name	Date modified	Type	Size
bin	9/23/2025 12:12 AM	File folder	
Settings	9/22/2025 11:42 PM	File folder	
EchoSigh-Pro Release Notes.pdf	9/23/2025 12:04 AM	PDF File	120 KB
EchoSight-Pro User Guidance.pdf	9/23/2025 1:12 AM	PDF File	3,389 KB
Start EchoSight-Pro.bat	7/24/2025 6:15 PM	Windows Batch File	1 KB

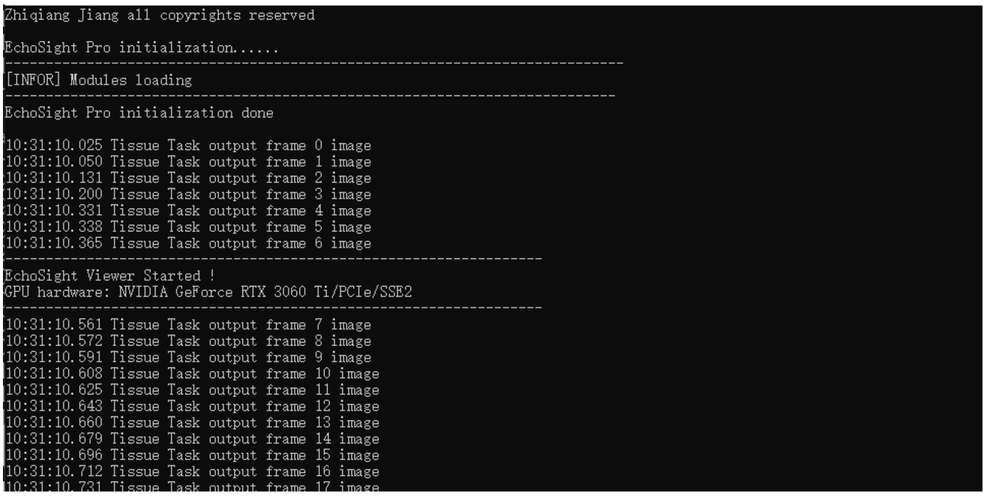
You can also double click **EchoSight-Pro.exe** in bin folder to run the program



1.2 Basic Usage Tricks

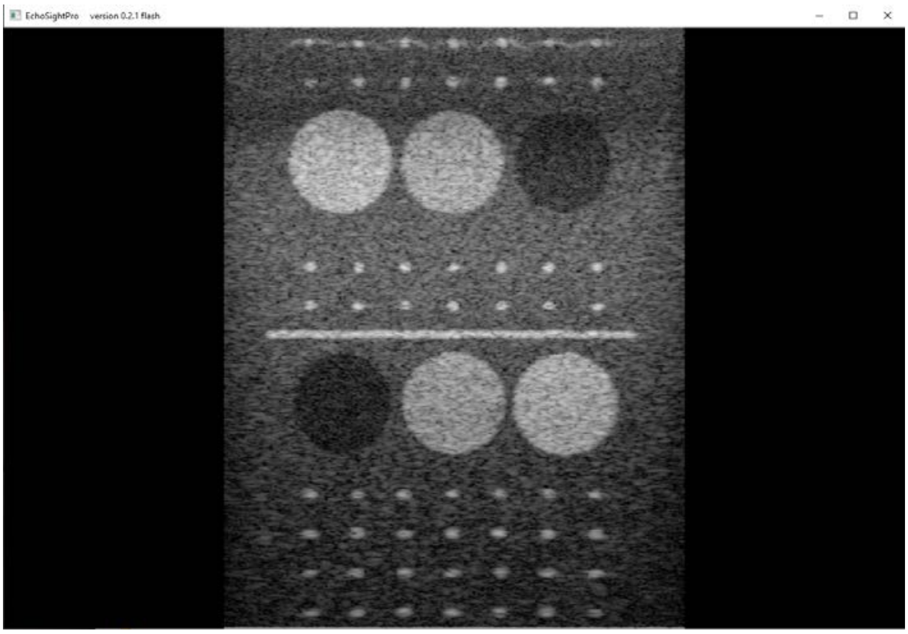
When EchoSight-Pro is running, it will use the full computation power based on default settings. It will pop out two windows, one is command line window, the other is a GUI window (EchoSight Viewer).

Command Line Window



Command line window will give the actual EchoSight-Pro processing status information, including what it is doing right now, as well as the actual speed it is working.

GUI Window



GUI window displays the final results after the whole processing. **Notice**, **the GUI window update rate is not the real processing speed**, you can check the real processing speed at background command line window with accurate time tags.

Important notice, when EchoSight-Pro is running the CPU or GPU is quite busy, if you do not want to burn it out (I'm joking).

Pause the EchoSight-Pro

Use your mouse to left click in the command line window, the whole EchoSight-Pro processing chain will pause and wait.

Resume/Continue EchoSight-Pro

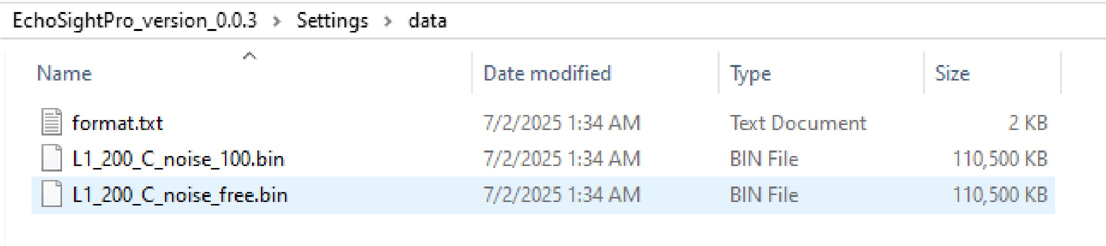
Press enter key in command line window, the EchoSight-Pro will resume from the point it paused at.

2. Data and Configuration

2.1 Raw Data File

2.1.1 file location and format

The EchoSight-Pro uses **raw channel IQ data** as its input. EchoSight-Pro will load in the raw data file from folder **EchoSightPro\Settings\data\XXX.bin**

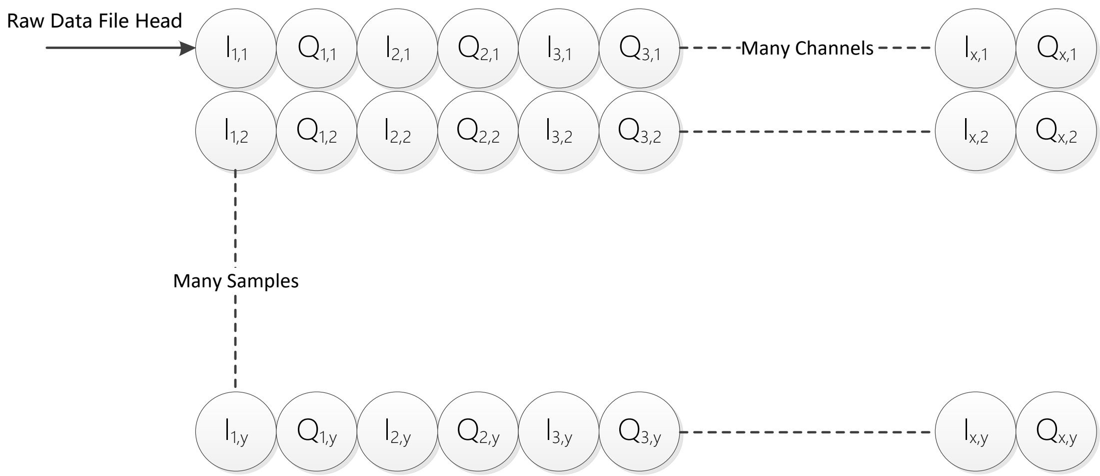


The raw channel data is too huge, I only put 2 basic data in the folder, you can simulate your own channel data based on all kinds of existing ultrasound simulation tools, or capture the real ultrasound channel data on your prototype machine and stores them in file.

To make EchoSight-Pro accept the correct data, the raw data format should follow the format as below:

1. **The data file is stored in raw binary format (not text!!!)**
2. **The data file is stored in the following order**

Format diagram



3. **The data sample is in 16 bits signed integer for each I, Q samples**

You can also know more details by checking the “format.txt” file in the data folder.

PS: Z.Jiang explanation for the data format
The EchoSight-Pro data file storing format is to make the FPGA comfortable after AFE fixed demodulation, and 16 bits integer is the majority AFE default sampling output (in majority cases, it might be 12 bits for 50MHz or 14 bits for 40MHz in reality, except some special chips)

2.1.2 get the raw data file

I have put a basic sample dataset file in the folder within the software package already. You can use it directly. But actually, you should obtain the data, specify the data to your specific needs.

1. Simulation data

There are many well-known tools for this purpose, such as Field 2, Focus, KWave, Must and etc. Their download link listed below
-Field II https://field-ii.dk/
-Focus https://www.egr.msu.edu/~fultras-web/
-MUST https://www.biomecardio.com/MUST/
-KWave http://www.k-wave.org/index.php

After you obtained the data, please reformat the data file in the format described in section 2.1.1.

2. Real Captured data

To use real data is cooler, right? You can use real data as well, EchoSight-Pro do not differentiate them at all.
Some commercial products provide raw data capture before beam forming stage, including but not limited to Versonics, Ultrasonix and some other cooperations**.**

After you captured the data, don't forget the step of transfer the data format as section 2.1.1 requires as well.

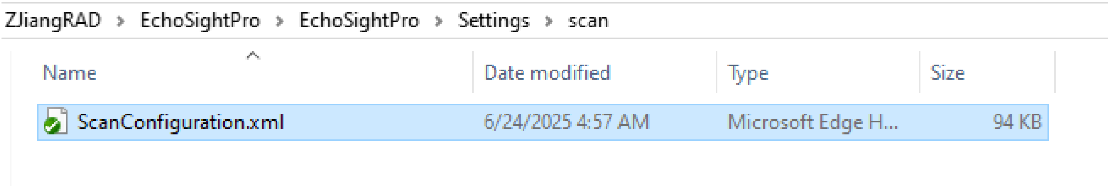
3. Data file-based VS Online Scan

No difference for EchoSight-Pro, the computation, data transfer work load are extremely similar. The file data transfer module is repeating copying the source memory data and doing the transfer instead of using the same data pointer without doing transferring.

2.2 Raw Data Settings

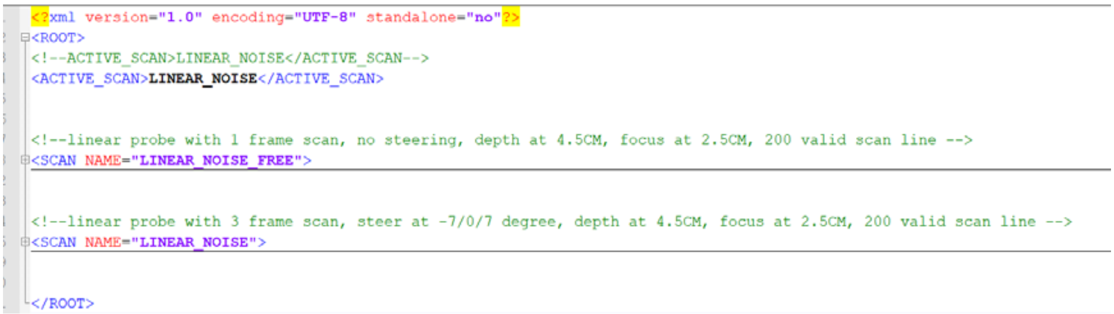
2.2.1 Raw Data Setting file

After you obtain the data already, you left one step ahead to make it work. Set the scan and hardware setting for EchoSight-Pro.
You can find the setting file from folder EchoSightPro\Settings\scan\



Open the XML setting file named **ScanConfiguration.xml** to define your personal needs (Strongly suggest you use NOTEPAD++ to edit the file)

It should look as below



2.2.2 Setting file item details

There might be one or more scans listed in the file, ignore them.
Focus on the **ACTIVE_SCAN** item in the root node. It describes which scan setting EchoSight-Pro will use.

Step 1: Create Your Scan
You should create your own SCAN settings. You can start with by copy paste the whole LINEAR_STD_1 context and rename it to your preferred name, let say “MY_COOL_SCAN”.

Step 2: Active Your Scan
You need to change the ACTIVE_SCAN content to “MY_COOL_SCAN”, otherwise it will still use the default one.

Step 3: Customize Your Hardware Setting
You need define your hardware settings according to the data you captured or simulated. There many items below, but for version 0.1.0 only the following ones are allowed modifications, which I marked blue ball by its left.

```xml
<SCAN NAME="LINEAR_STD_1">
    <PROBE_TYPE>LINEAR</PROBE_TYPE>
    <PROBE_RADIUS_MM>0</PROBE_RADIUS_MM>
    <PROBE_HALF_ANGLE>0</PROBE_HALF_ANGLE>      <!--unit shou
    <ELE_NUM>128</ELE_NUM>
    <RCV_CH_NUM>128</RCV_CH_NUM>
    <RCV_CH_MIN_NUM>6</RCV_CH_MIN_NUM>
    <PITCH_MM>0.3</PITCH_MM>
    <!--USED FOR CUTOFF AND pos_x_mING REFRACTION UNIT:MM, fc
    <LEN_THICK_MM>0</LEN_THICK_MM>
    <!---Micro second is enough for our simulator, nano secor
    <SCAN_TIME_MICRO_SEC>1000</SCAN_TIME_MICRO_SEC>
    <FULL_PULSE_HDTIME_NS>1366</FULL_PULSE_HDTIME_NS>
    <!--TX PULSE HEADING Time of transmit pulse unit:nano sec
    <TX_PULSE_HDTIME_NS>0</TX_PULSE_HDTIME_NS>
    <DATA_SRC>LOAD_FILE</DATA_SRC>          <!--LOAD_FILE:loa
    <DATA_NOISE_MAG>5</DATA_NOISE_MAG>                  <!--
    <!--data file size proximate estimation: file size = 2 *
    <DATA_FILES>L1_200_C.bin</DATA_FILES>   <!-- for multiple
    <!---Uniform steer settings for all. Unit degree,negative
    <SCAN_SEQ_STEER>0</SCAN_SEQ_STEER>      <!-- for multiple
    <UserMode>B Fund</UserMode> <!-- Options:B Fund,B Fund Cc
```

EIE_NUM: this is the element number of your probe.
RCV_CH_NUM: this defines your machine channel number, associate with the raw data format.
PITCH_MM: this declares your probe geometry element pitch size, unit in mm.
DATA_FILES: this let EchoSight-Pro to know which raw file to load in.
(for version 0.1.0 only 1 file is allowed)

Step 4: Customize Your Scan Group

Scan group defines more details of your scan, one scan group corresponding to raw data file, in the order of the file names you give in the DATA_FILES. (version 0.1.0 only provide 1 file anyway)

```xml
    <SCAN_GROUP ID = "1">
        <!---SYN_SHIFT_NUM works in channel synthetic mo
        <SYN_SHIFT_NUM>0</SYN_SHIFT_NUM>
        <MLA_NUM>1</MLA_NUM>
        <SYN_NUM>1</SYN_NUM>
        <!--HAN for hanning window, HAM for hamming wind
        <RX_APO>HAN</RX_APO>
        <RX_FN>0.7</RX_FN>
        <!---TX_TYPE can only be: 1.FOCUSED_WAVE 2.DIVER
        <TX_TYPE>FOCUSED</TX_TYPE>
        <TX_FREQ>7100000</TX_FREQ>
        <!--AFE fixed demodulation frequency-->
        <AFE_FD>7100000</AFE_FD>
        <IQ_RATE>12500000</IQ_RATE>
        <ACQ_DEPTH_MM>55</ACQ_DEPTH_MM>
        <TX_FOCUS_MM>25</TX_FOCUS_MM>
        <SOUND_SPEED>1545</SOUND_SPEED> <!---UNIT: meter
        <TX_NUM>200</TX_NUM>
```

MLA_NUM defines how many parallel lines you want to beam form within beam formation stage;
RX_FN defines the receiving F# value;

AFE_FD: defines the fix demodulation frequency AFE chip does. If you obtain the data from simulation, you should modulate it yourself from RF to IQ and gives the AFE_FD value here.

IQ_RATE: defines your AFE output data sample rate after down sampling, you should give its corresponding value here. If you obtain the data from simulation, you can down sample on your own and gives the value here.

ACQ_DEPTH_MM: your actual scan beam depth.
TX_FOCUS_MM: your actual transmition focus depth, unit in mm. You can set it within the FOV or out of FOV, or quite far away (let say 999999999) for the plane wave case.

SOUND_SPEED: the sound speed value you prefer for beam formation, it will impact the beam forming output.

TX_NUM: this should match the number of transmit line number.

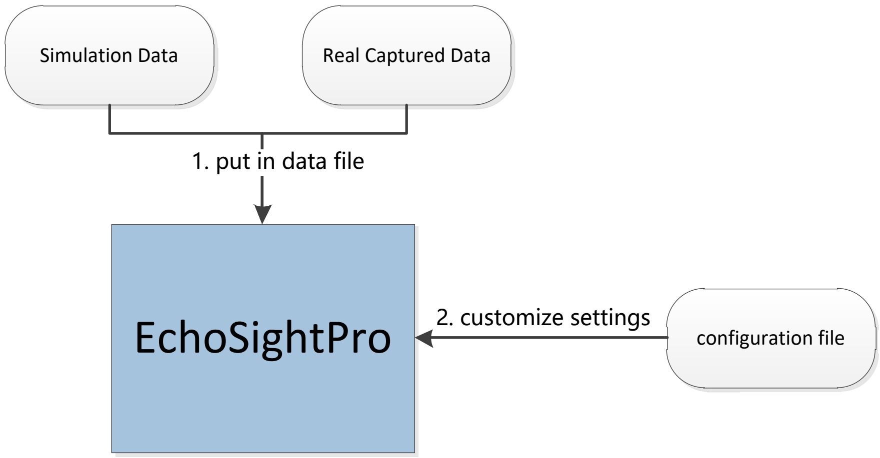
Step 5: Customize Your Scan Sequence
This is the last step.

```
<SCAN_SEQ>
    <!---SCAN INSTRUCTION DEFINITION 3 TYPES: 1.VALID,2.INVALID,3.DUMMY-->
    <TX_ITEM id="0" pos_x_m="-0.01920" steer = "0.0" inversion = "0" mode = "B" frame_id = "1">valid</TX_ITEM>
    <TX_ITEM id="1" pos_x_m="-0.01901" steer = "0.0" inversion = "0" mode = "B" frame_id = "1">valid</TX_ITEM>
    <TX_ITEM id="2" pos_x_m="-0.01881" steer = "0.0" inversion = "0" mode = "B" frame_id = "1">valid</TX_ITEM>
    <TX_ITEM id="3" pos_x_m="-0.01862" steer = "0.0" inversion = "0" mode = "B" frame_id = "1">valid</TX_ITEM>
```

Each tx item defines the property of actual transmit.
Each tx item number should matches the TX_NUM you give above. The pos_x and steer define the transmit kick off position in space. pos_x is in meter unit; steer gives the steering angle in radian unit.

By now, EchoSight-Pro gets everything needed to run.

From my perspective, I think it’s quite easy.


Simulation Data
Real Captured Data
1. put in data file
EchoSightPro
2. customize settings
configuration file

3. Investigate by Examples

3.1 Kick Off with EchoSight-Pro

Open the “ScanCongiguration.xml” file to confirm the suitable scan set is selected and correct raw data file is chosen.

EchoSightPro › Settings › scan

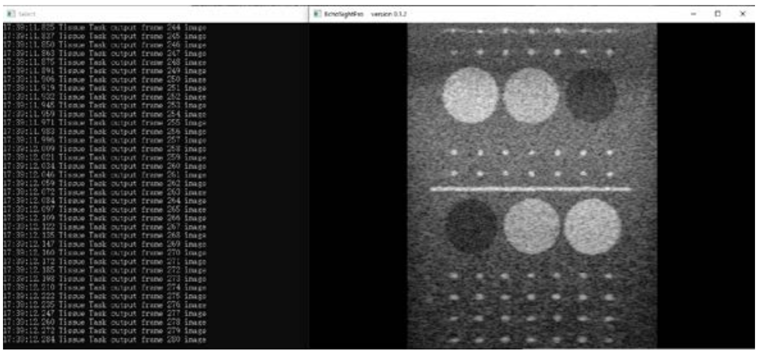
Name	Date modified	Type	Size
ScanConfiguration.xml	7/1/2025 11:06 PM	Microsoft Edge H...	50 KB

For version 0.1.0, please choose LINEAR_BASIC scan set. Make sure “DATA_SRC” is LOAD_FILE, and corresponding “DATA_FILES” is set as “L1_200_C_noise_free.bin”

```
<ROOT>
<ACTIVE_SCAN>LINEAR_BASIC</ACTIVE_SCAN>

<!--linear probe with 1 frame scan, no steering, depth at 4.5CM, focus at 2.5CM, 200 valid scan line -->
<SCAN NAME="LINEAR_BASIC">
    <PROBE_TYPE>LINEAR</PROBE_TYPE>
    <PROBE_RADIUS_MM>0</PROBE_RADIUS_MM>
    <PROBE_HALF_ANGLE>0</PROBE_HALF_ANGLE>      <!--unit should be radian, data type float-->
    <ELE_NUM>128</ELE_NUM>
    <RCV_CH_NUM>128</RCV_CH_NUM>
    <RCV_CH_MIN_NUM>6</RCV_CH_MIN_NUM>
    <PITCH_MM>0.3</PITCH_MM>
    <!--USED FOR CUTOFF AND pos_x_mING REFRACTION UNIT:MM, for real scan check the probe spec, for simulation le
    <LEN_THICK_MM>0</LEN_THICK_MM>
    <!---Micro second is enough for our simulator, nano second is not reliable on OS either.This will control s
    <SCAN_TIME_MICRO_SEC>1000</SCAN_TIME_MICRO_SEC>
    <FULL_PULSE_HDTIME_NS>1366</FULL_PULSE_HDTIME_NS>
    <!--TX PULSE HEADING Time of transmit pulse unit:nano second, this works for the HW and simulation cases bo
    <TX_PULSE_HDTIME_NS>0</TX_PULSE_HDTIME_NS>
    <DATA_SRC>LOAD_FILE</DATA_SRC>          <!--LOAD_FILE:load from file for simulator,GEN_DATA:no file needed,
    <DATA_NOISE_MAG>5</DATA_NOISE_MAG>              <!-- Random noise level range numerical value generated
    <!--data file size proximate estimation: file size = 2 * acquisition depth / sound speed * IQ_RATE * REV CH
    <DATA_FILES>L1_200_C_noise_free.bin</DATA_FILES>    <!-- for multiple inputs, separate by ";"-->
```

L1_200_C_noise_free.bin and L1_200_C_noise_100.bin are two files, I simulated for EchoSight-Pro version 0.1.2 usage, and stored in EchoSightPro\Settings\data folder.

After all the confirmation, click “Start EchoSight-Pro.bat”
You should see the output below



3.2 Modify the Settings

You can change the settings as you like, the beam formed image quality will be impacted as well. Open the “ScanCongiguration.xml” file again, modify the “**SOUND_SPEED**” item under “SCAN_GROUP” in “LINEAR_BASIC” scan settings.

```xml
<SCAN_GROUP ID = "1">
    <!---SYN_SHIFT_NUM works in channel synthetic mode, if num = 0, means no synthetic at all, s
    <SYN_SHIFT_NUM>0</SYN_SHIFT_NUM>
    <MLA_NUM>1</MLA_NUM>
    <SYN_NUM>1</SYN_NUM>
    <!--HAN for hanning window, HAM for hamming window,BLACKMAN for black man window, COS for cc
    <RX_APO>HAN</RX_APO>
    <TX_FN>1.2</TX_FN>
    <RX_FN>0.7</RX_FN>
    <!---TX_TYPE can only be: 1.FOCUSED_WAVE 2.DIVERSED_WAVE 3.PLANE_WAVE-->
    <TX_TYPE>FOCUSED</TX_TYPE>
    <TX_FREQ>7100000</TX_FREQ>
    <!--AFE fixed demodulation frequency-->
    <AFE_FD>7100000</AFE_FD>
    <IQ_RATE>10000000</IQ_RATE>
    <ACQ_DEPTH_MM>50</ACQ_DEPTH_MM>
    <TX_FOCUS_MM>25</TX_FOCUS_MM>
    <SOUND_SPEED>1640</SOUND_SPEED> <!---UNIT: meters per second, it impact not beam formation c
    <TX_NUM>200</TX_NUM>
    <!---Uniform steer settings for all. Unit degree,negative stands for left steer,positive rep
    <SCAN_SEQ_STEER>0.0</SCAN_SEQ_STEER>
    <!---MODE can only be: B,B_Hfilter,B_inversion,C,PW,CW,Const,ConstAdv,Elasto,ShWave etc.-->
```

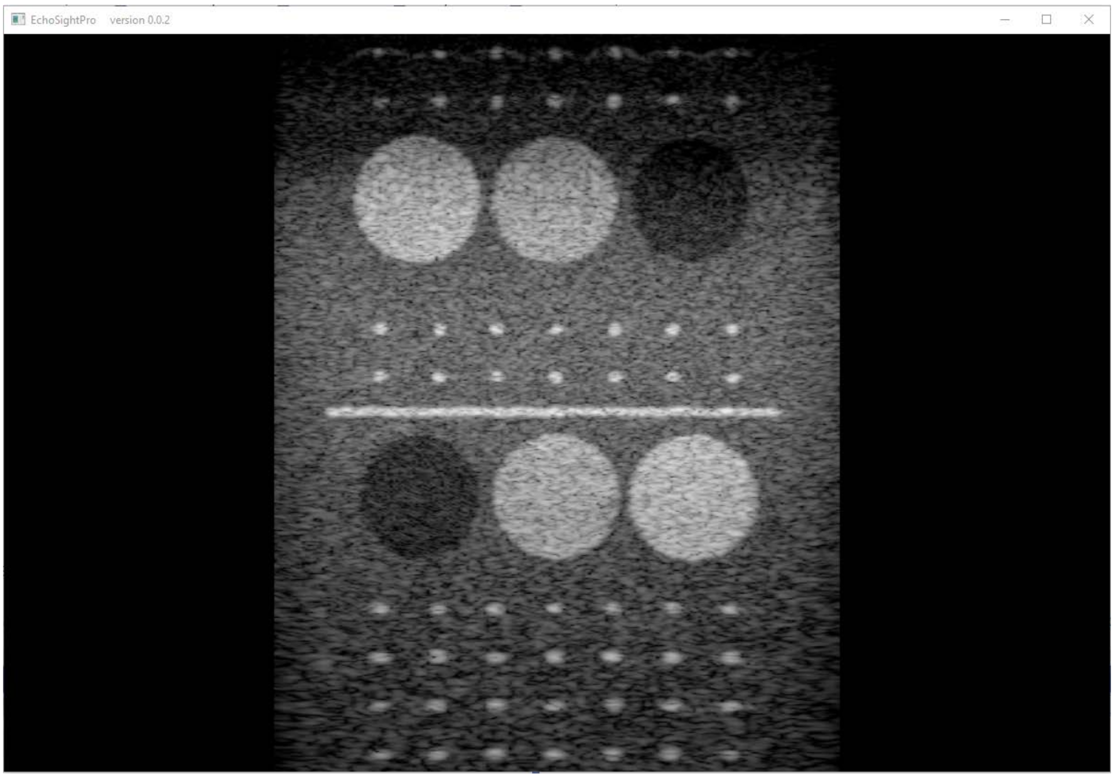
Then double click “Start EchoSight-Pro.bat” to EchoSight-Pro to see what will happen.
You can also change other items such as “**RX_FN**”, “**RX_APO**” , “**RX_FN**” to see what will happen.

3.3 Have MLA and RTB

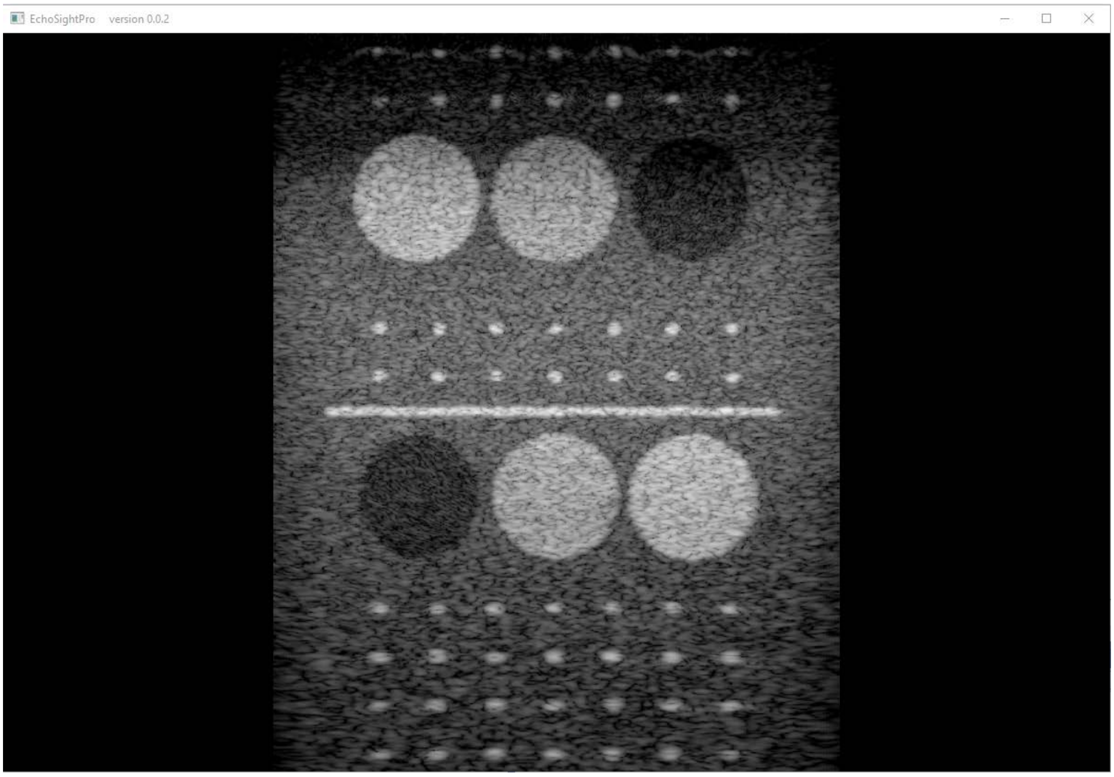
MLA is one of most important parameters in settings. It orders EchoSight-Pro to beam form multiple lines with 1 fire channel data.

```xml
<SCAN_GROUP ID = "1">
    <!---SYN_SHIFT_NUM works in channel synthetic mode, if num = 0, means no synthetic at all, sec
    <SYN_SHIFT_NUM>0</SYN_SHIFT_NUM>
    <MLA_NUM>5</MLA_NUM>
    <SYN_NUM>1</SYN_NUM>
    <!--HAN for hanning window, HAM for hamming window,BLACKMAN for black man window, COS for cosi
    <RX_APO>HAN</RX_APO>
```

When your MLA is bigger than 1, the RTB will enable automatically as well. **Both MLA and RTB are built in EchoSight-Pro already and work in real time**. Try it to see what will happen. Two Images are shown below, both of them are of 200 transmissions, but the first image is of 1 MLA while the second image is of MLA 4, which is 4 times line density along horizontal direction. Check the difference below



Transmit 200 times, with MLA 1



Transmit 200 times, with MLA 4

Look at how much horizontal resolution we have obtained by higher MLA!
RTB attempts to fix the sound wave front position error by shifting A/D sample position after

receiving the signal. Although it cannot completely fix that, it does fix majority of errors in wave front position mis-match.
Turn on MLA will increase the EchoSight-Pro processing working load obviously.
Let say your data rate is 5 Giga Bytes per second, MLA 20 means EchoSight-Pro needs to process 5 Giga Bytes data calculation 20 times per second, which means 100 Giga Bytes of data computation per second on your computer.
It's quite a lot data, baby. That won't be a small challenge for your computer and Z.Jiang's system engineering design capability as well.

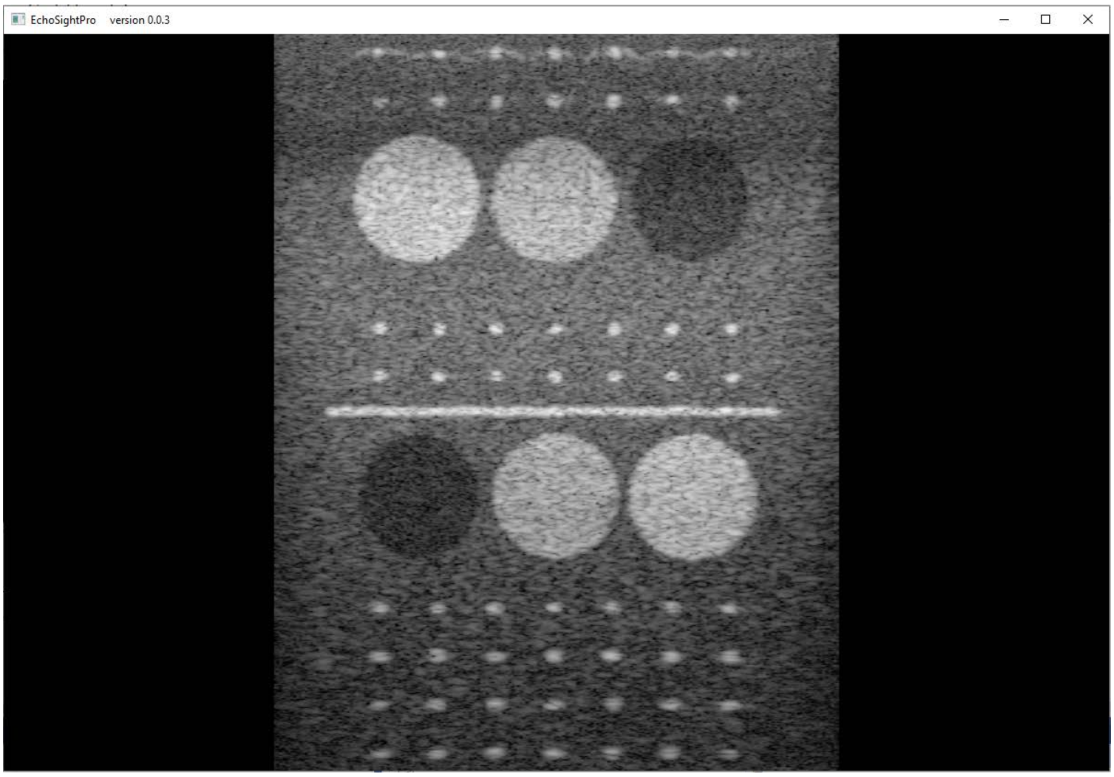
3.4 Synthetic Now

Synthetic aperture is a traditional technique widely used in radar, sonar, ultrasound system design. **EchoSight-Pro has built in real time synthetic technique already**. But remember to turn on the multiple line acquisition beforehand, otherwise we do NOT have multiple data to synthetic image at all.

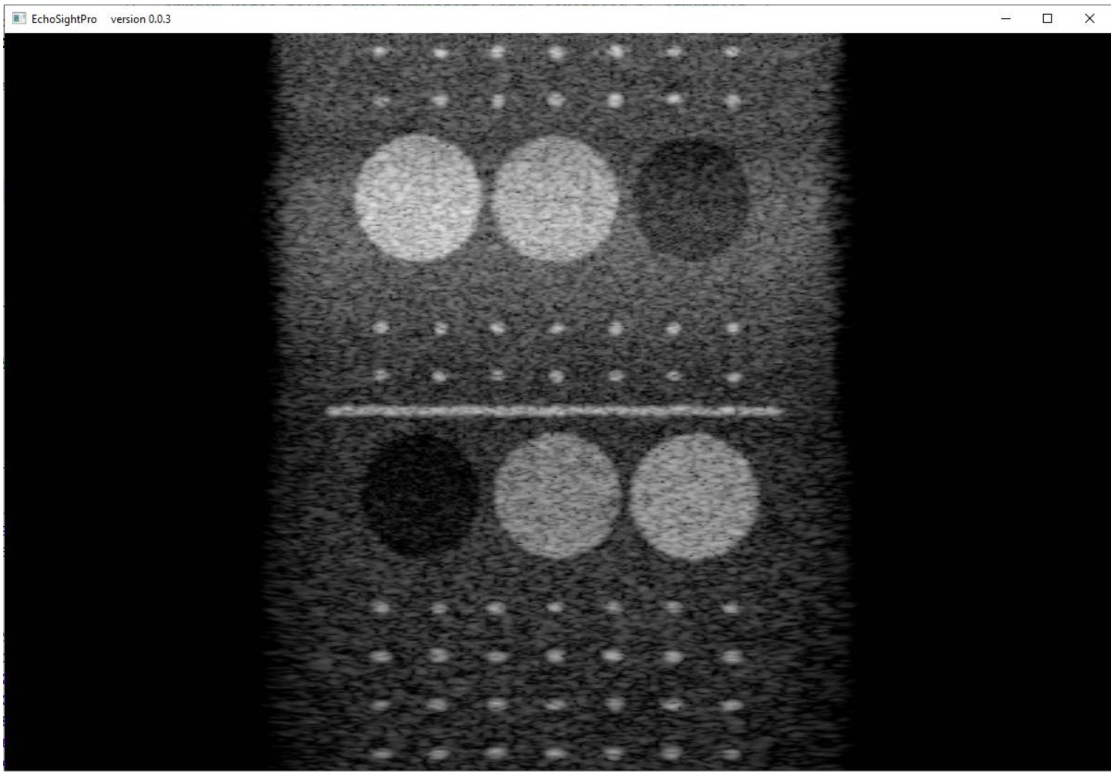
```
<MLA_NUM>8</MLA_NUM>
<SYN_NUM>8</SYN_NUM>
```

Check the two images generated by EchoSight-Pro below, which generated from the identical channel data where I added in value 100 random noises in the channel data. (please use **L1_200_C_noise_100.bin** instead of **L1_200_C_noise_free.bin**)

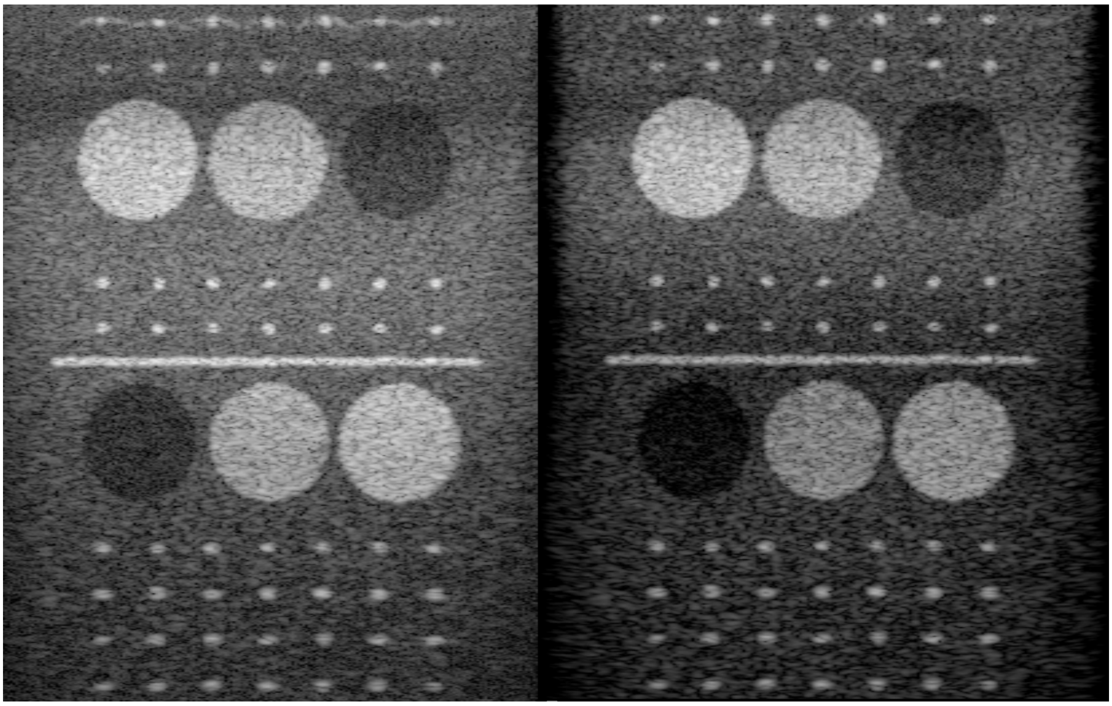
The first image is of MLA 1, SYN 1(synthetic 1 time means no synthetic at all)



Check the second image for comparison, which is of MLA 24 SYN 24.



I put them side by side below, it will be more obvious



The IQ improvement is dramatic, mainly on 3 facets.

1. Point becomes tighter, grating lobe drops.

This is mainly caused by synthetic, and also benefits from RTB (EchoSight-Pro build in RTB and works automatically) some.

2. The penetration (SNR) improves at far fields a lot.

This is also caused by synthetic; SNR will increase 3dB every time when you double your synthetic number. This makes signal pops out while noise drops down, which leads to signal penetration improvement.

3. Contrast resolution increases a lot.

Contrast resolution is an important IQ factor, which was introduced by Legendary ultrasound company ACCUSON. It is used to evaluate the machine's capability to differentiate different region.

Synthetic combination of signal will lead to overall signal drop a little, but darker area drops more, which leads to the difference between the white and dark area increase in return. This makes machine's contrast resolution increase. (Please check the cyst differentiation between the background)

3.5 Post params

Majority of the parameters are not suggested to change, except those.

In the task\ EchoSight_Tissue_Task.xml file

PRE_GAIN and **POST_GAIN** should be configured according to your actual data. Please check the description comment I gave.

```xml
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<ROOT>

<TASK>
    <NAME>EchoSight_Tissue_Task</NAME>
    <HEAD_TASK>FALSE</HEAD_TASK>
    <INPUT_ScanDataType>B|B_inversion</INPUT_ScanDataType><!-- valid value includes: B,B_Hfilter,B_inversion,C,Pw_basic,Any -->
    <LINE_INTERP>NONE</LINE_INTERP>      <!--optional value: NONE,COMP,ENV-->
    <INTERP_ALG>LIN</INTERP_ALG>            <!--optional value: LIN,NLIN,CUB-->
    <MEM_FORCE>0</MEM_FORCE>             <!--1 to N defines the memory worker. It should be optimized for each task-->
    <CALC_FORCE>2</CALC_FORCE>           <!--1 to N defines the calculation force. It should be optimized for each task-->
    <THREAD_NUM>2</THREAD_NUM>           <!--Not too many threads please, that would NOT help-->
    <PRE_GAIN>7.2</PRE_GAIN>             <!--pre gain unit in dB, using float value, suggested range between [0 30]. If surpassed
    <POST_GAIN>-135</POST_GAIN>          <!--post gain should always below zero range between [-255 0] for image value, actually,
    <MAX_BUFFER_NUM>5</MAX_BUFFER_NUM><!--max buffer number should be positive and optimized-->
</TASK>

</ROOT>
```

3.6 Others

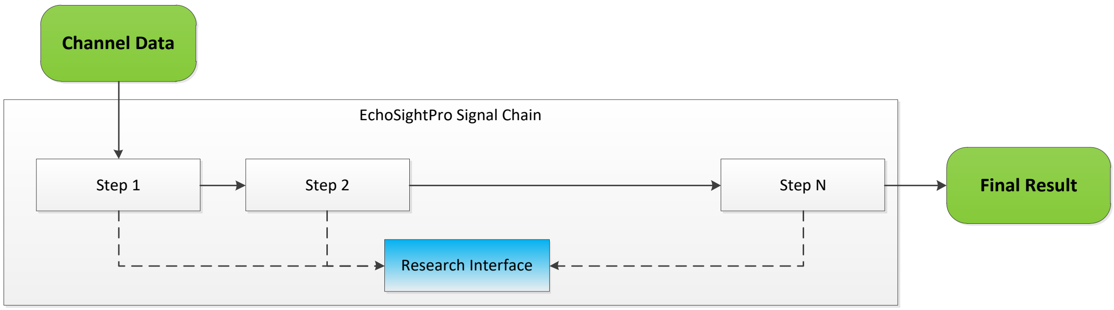
TBD

4. Research Interface

For academic research convenience, a simple research interface module is added in EchoSight-Pro since version 0.5.1. Through this research interface, you can do your personalized process, algorithm evaluation within EchoSight-Pro signal chain.

4.1 Research Interface Introduction

Research Interface module is a simple DLL, which will be called within EchoSight-Pro signal chain. You can take the data output and doing your personalized process, and set it back to the signal chain again, if you want.



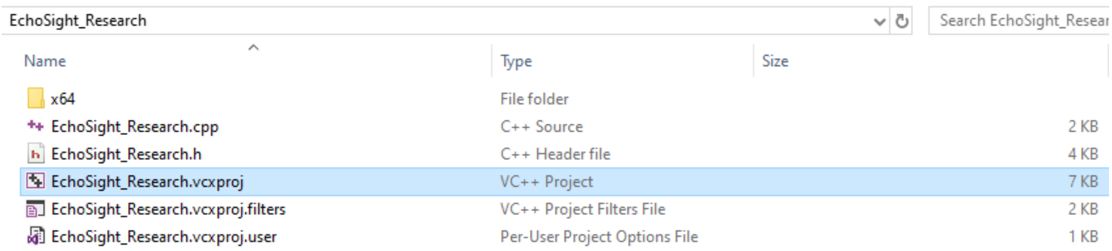
It's simple, but very handy for variant research purpose.

Research interface module is provided with source codes in the EchoSight-Pro software package. You can any modifications to meet your needs.

4.2 Interface Usage Example

If you want to do some image processing, let say you want "right half image gray scale value to be reverted".

- **Step 1** Open the "**EchoSight_Research**" project



- **Step 2** Modify the source codes in the CPP file according to your needs.

(There are useful comments within the codes, please follow the guidance of the comments)

For our example, we make the master flag and image process flag enabled first.

```
EchoSight_Research::EchoSight_Research()
{
    //Z.Jiang memo: initialize your flag here first...
    m_master_flag = true;

    m_rtbProcess_flag = false;
    m_synProcess_flag = false;
    m_imgProcess_flag = true;
}
```

Modify the image process function according to your needs.

```
void EchoSight_Research::Func_Img(unsigned char* _in_out_line_data, const size_t sample_num, const size_t _line_id, const size_t _line_num)
{
    //input data is right after log compression in the form of 8bit unsigned gray image
    //do your personalized process on image data here...
    if (_line_id > (_line_num >> 1))
    {
        for (size_t sample_id = 0; sample_id < sample_num; sample_id++)
            _in_out_line_data[sample_id] = 0xff - _in_out_line_data[sample_id];
    }
}
```

- **Step 3** Build the project to generate the DLL and LIB files

x64 › Release | Search Release

Name	Type	Size
EchoSight_Research.dll	Application extension	12 KB
EchoSight_Research.exp	Exports Library File	4 KB
EchoSight_Research.lib	Object File Library	7 KB
EchoSight_Research.pdb	Program Debug Database	444 KB

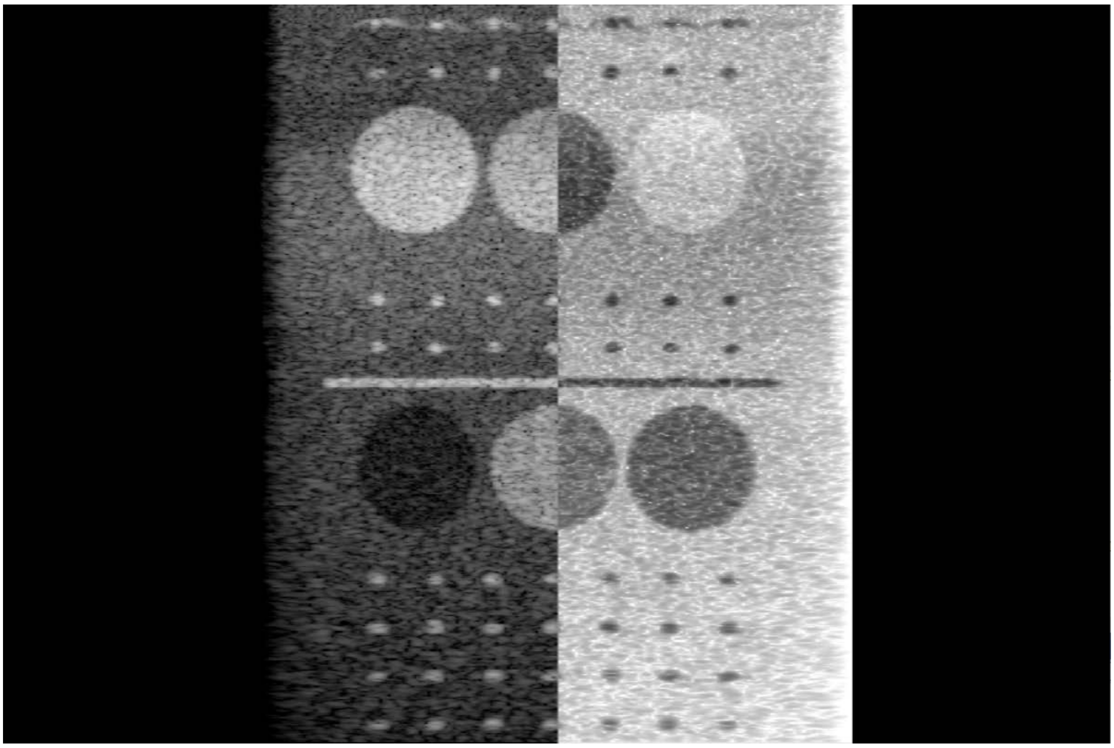
- **Step 4** Copy “EchoSight_Research.dll” file to the bin folder of EchoSight-Pro package, replace the existing one.

bin

Name	Type	Size
EchoSight-Pro-flash.exe	Application	33 KB
Alg_Tissue.dll	Application exten...	31 KB
CL_Util.dll	Application exten...	20 KB
Common.dll	Application exten...	39 KB
Echo_Apo.dll	Application exten...	55 KB
Echo_Force.dll	Application exten...	100 KB
Echo_RTB_Delay.dll	Application exten...	61 KB
Echo_SoundPressure.dll	Application exten...	38 KB
EchoSight_Color_Task.dll	Application exten...	70 KB
EchoSight_Outport_Task.dll	Application exten...	77 KB
EchoSight_Research.dll	Application exten...	12 KB

After all steps done, your personalized process will be added into the EchoSight-Pro chain.

In our simple example, you will see the result as below.



In order to meet variant research needs, the IQ data interface after RTB or synthetic aperture is provided as well.

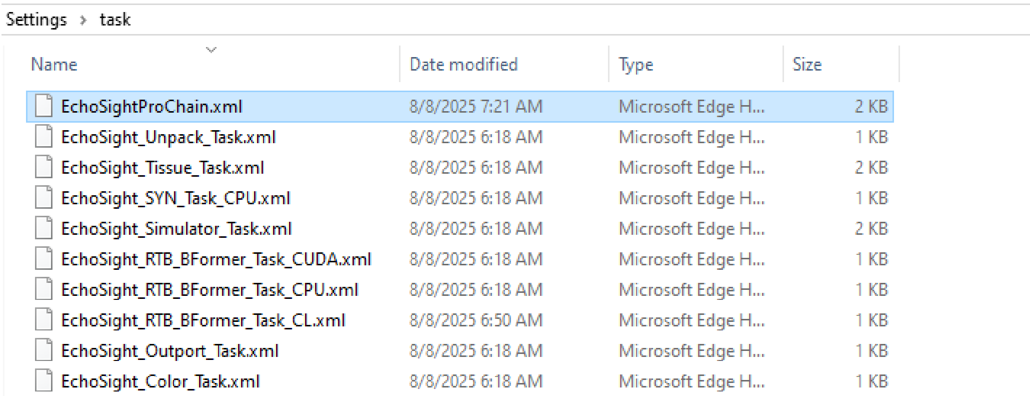
```
void EchoSight_Research::Func_RTB_IQ(float* _in_out_line_data, const size_t sample_num, const size_t _line_id, const size_t _line_num)
{
    //input data is right after recursive transmit beam forming
    //do your personalized process on iq data here...
}

void EchoSight_Research::Func_SYN_IQ(float* _in_out_line_data, const size_t sample_num, const size_t _line_id, const size_t _line_num)
{
    //input data is right after synthetic aperture processing
    //do your personalized process on iq data here...
}
```

5. CPU/Integrated GPU settings

5.1 CPU/Integrated GPU work mode switch

Since **version 0.3.0** or later, EchoSight-Pro providing different working mode. You can let beam forming work on **CPU mode**, or **integrated GPU mode** through configuration.

Find the "**EchoSightProChain.xml**" file under folder Settings/task shown below


Settings › task

Name	Date modified	Type	Size
EchoSightProChain.xml	8/8/2025 7:21 AM	Microsoft Edge H...	2 KB
EchoSight_Unpack_Task.xml	8/8/2025 6:18 AM	Microsoft Edge H...	1 KB
EchoSight_Tissue_Task.xml	8/8/2025 6:18 AM	Microsoft Edge H...	2 KB
EchoSight_SYN_Task_CPU.xml	8/8/2025 6:18 AM	Microsoft Edge H...	1 KB
EchoSight_Simulator_Task.xml	8/8/2025 6:18 AM	Microsoft Edge H...	2 KB
EchoSight_RTB_BFormer_Task_CUDA.xml	8/8/2025 6:18 AM	Microsoft Edge H...	1 KB
EchoSight_RTB_BFormer_Task_CPU.xml	8/8/2025 6:18 AM	Microsoft Edge H...	1 KB
EchoSight_RTB_BFormer_Task_CL.xml	8/8/2025 6:50 AM	Microsoft Edge H...	1 KB
EchoSight_Outport_Task.xml	8/8/2025 6:18 AM	Microsoft Edge H...	1 KB
EchoSight_Color_Task.xml	8/8/2025 6:18 AM	Microsoft Edge H...	1 KB

You will find the "**ACTIVE_TASK_CHAIN**" item on the top. Within this item you can configure which setting you prefer. The option settings are listed below, CPU_CHAIN and CL_CHAIN. When you set it as CPU_CHAIN.

```
<ROOT>
<!--ACTIVE_TASK_CHAIN ECHO_SIGHT_PRO_MODE="OFF_LINE">CL_CHAIN</ACTIVE_TASK_CHAIN-->
<ACTIVE_TASK_CHAIN ECHO_SIGHT_PRO_MODE="OFF_LINE">CPU_CHAIN</ACTIVE_TASK_CHAIN>

<!--Z.Jiang memo: It is an important system design path for low cost system design.But still
<TASK_CHAIN NAME="CPU_CHAIN">
    <CHAIN>EchoSight_Simulator_Task|EchoSight_Unpack_Task|EchoSight_RTB_BFormer_Task_CPU|Echo
</TASK_CHAIN>

<!--Z.Jiang memo: It is the most critical system design path to cut the cost while maintain t
<TASK_CHAIN NAME="CL_CHAIN">
    <CHAIN>EchoSight_Simulator_Task|EchoSight_Unpack_Task|EchoSight_RTB_BFormer_Task_CL|EchoS
</TASK_CHAIN>
```

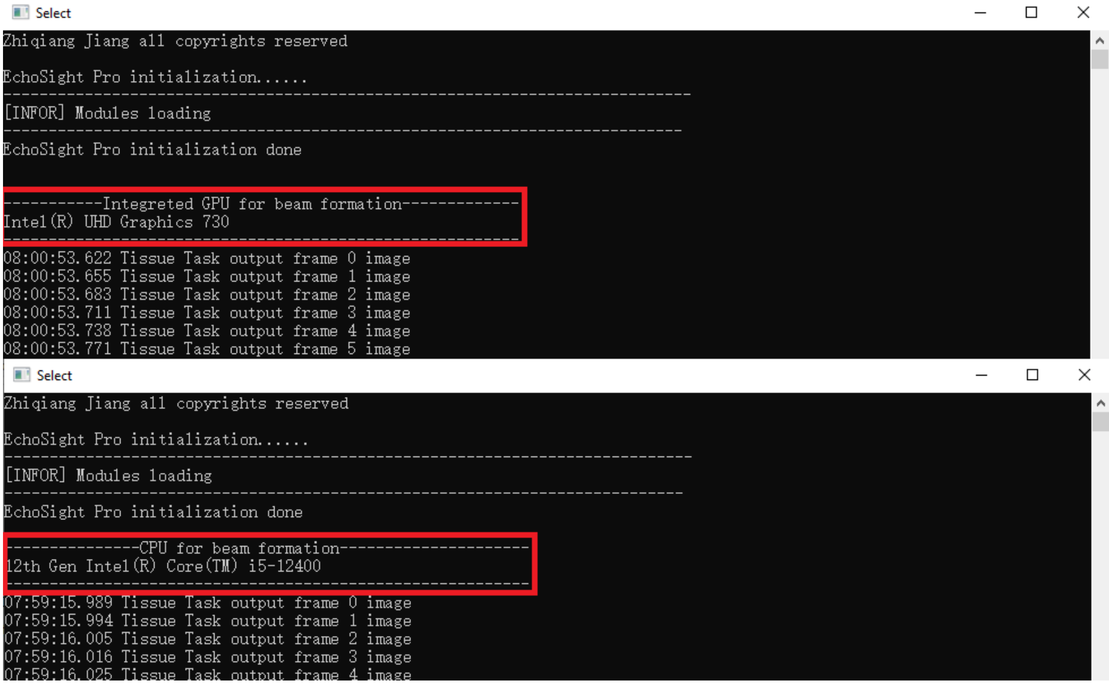
The beam forming work will use CPU for computation, if you set it as CL_CHAIN, it will put the work load on integrated GPU.

```
<ACTIVE_TASK_CHAIN ECHO_SIGHT_PRO_MODE="OFF_LINE">CL_CHAIN</ACTIVE_TASK_CHAIN>

<!--Z.Jiang memo: It is an important system design path for low cost system desig
<TASK_CHAIN NAME="CPU_CHAIN">
    <CHAIN>EchoSight_Simulator_Task|EchoSight_Unpack_Task|EchoSight_RTB_BFormer_Ta
</TASK_CHAIN>

<!--Z.Jiang memo: It is the most critical system design path to cut the cost whil
<TASK_CHAIN NAME="CL_CHAIN">
    <CHAIN>EchoSight_Simulator_Task|EchoSight_Unpack_Task|EchoSight_RTB_BFormer_Ta
</TASK_CHAIN>
```

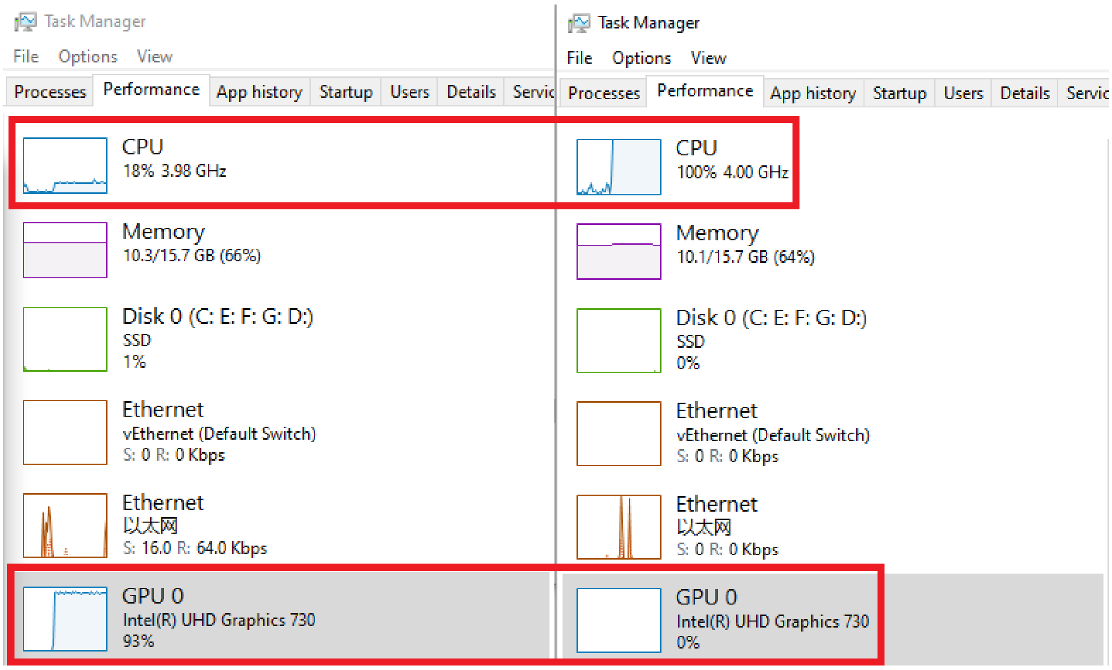
I have checked image correct on intel UHD 730 integrated GPU already. It should be workable on AMD integrated GPU also, but I haven't check yet, bugs may exist on AMD integrated GPU.

5.2 CPU/Integrated GPU work mode check

There are two ways to check the working mode, after you have made the configuration. One way is to check command window information. At the initialization stage, command window will show which hardware is used for beam forming, check the screen shot below.



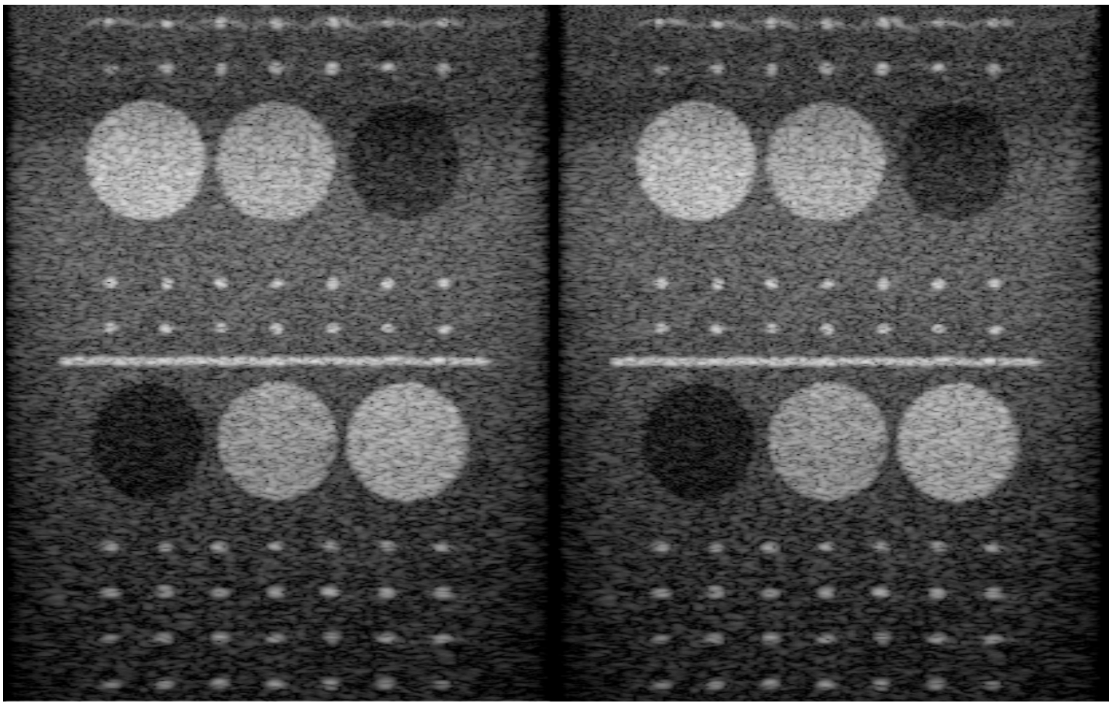
Another way is to check your windows task manager.



5.3 CPU/Integrated GPU image comparison

There is image difference between CPU and Integrated GPU setting due quantization errors or undiscovered bugs.

Following two images are generated with different hardware settings in version 0.4.0



(Left: CPU setting, Right: Integrated GPU setting)

6. Performance

6.1 Performance Description

Full chain data process rate is the **maximum** data input rate EchoSight-Pro can handle, it includes the whole process chain from getting channel data to final log compressed output. It is the **core indicator** for software beamforming-based ultrasound scanner.

Full chain maximum data rate should above **5.12GB/Sec for 128-channel system**
Full chain maximum data rate should above **2.56GB/Sec for 64-channel system**

6.2 Performance Summary

The performance of version 0.5.1 can **vary depends on your computer hardware** capabilities. My testing results are based on my personal computer hardware.

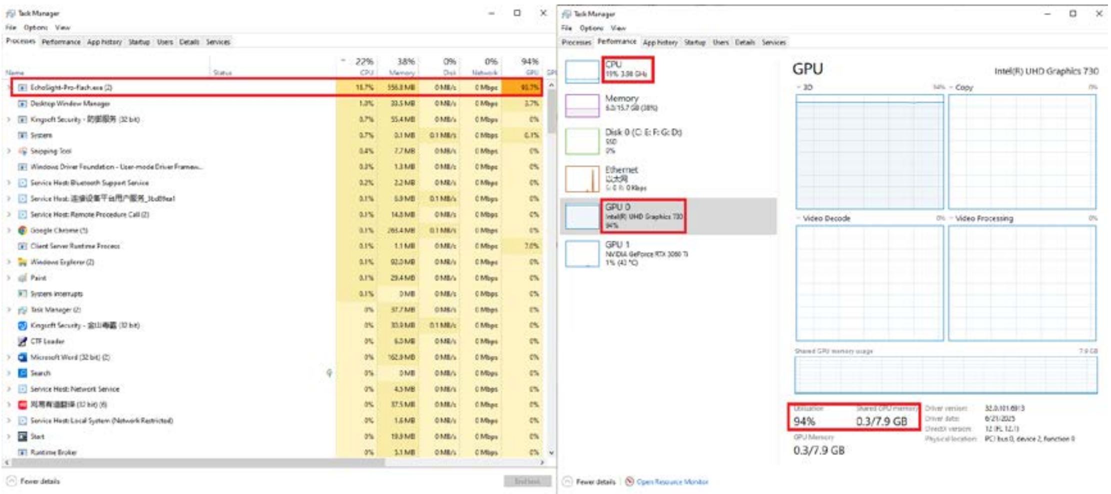
EchoSight-Pro v0.5.1 Data Chain Effciency Test Summary
(Unit: Giga Byte Per Second)

CPU		Memory	Integrated GPU
Gen 12th Intel i5-12400		16G @ 2400MHz DDR	Intel UHD730
MLA	**SYN**	**Intel I5-12400 CPU setting**	**Intel UHD730 Integrated GPU setting**
1	1	19.54	**23.44**
2	2	11.23	**24.30**
4	4	6.15	**23.65**
6	6	4.10	**14.90**
8	8	3.18	**14.72**
10	10	2.58	**10.45**
12	12	2.16	**10.36**
16	16	1.58	**8.06**
20	20	1.27	**6.41**
24	24	1.01	**5.29**
32	32	0.76	**3.81**

Notice: when data rate is too high (like above 20GB/sec), the data rate is becoming unstable from time to time. I do NOT want to pick good data to make it beautiful or reasonable. I just record what is monitored.

Given the same hardware limitation:
For CPU setting, it is the **fastest** in the world by now.
For integrated GPU setting, it is the **fastest** in the world after v0.5.0 by now.
(If being surpassed, please provide information to echosight_pro@126.com, I will correct my description mistakes.)

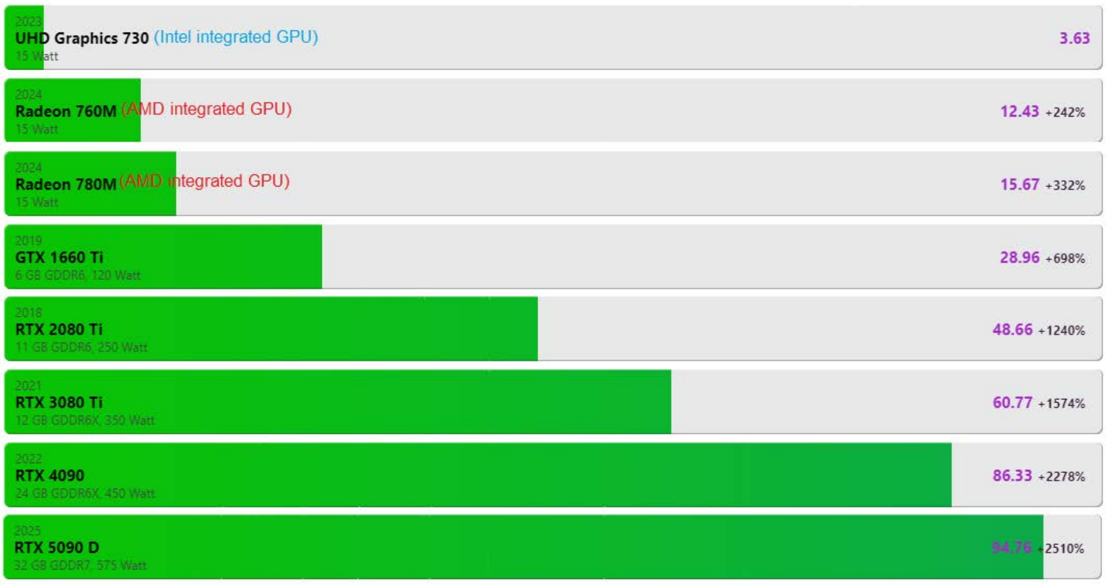
For the integrated GPU setting test, we can observe pretty low CPU loading, empty GPU loading on independent GPU (Nvidia hardware not used at all), pretty low memory foot print, only integrated GPU (intel UHD730) is quite busy on my computer.



6.3 Hardware benchmark

My development is based on Intel integrated GPU UHD730 and 2400MHz DDR memory.
It is **NOT** because it is friendly to R&D engineering, but because it is weak, **extremely weak**.

For GPU hardware capability:
For GPU overall hardware capability comparison with mainstream GPU, you can find hardware evaluations from the 3rd party such as:
Power Tech(https://www.techpowerup.com/), Technical City(https://technical.city/en);
UHD 730 computation capability comparison.



(UHD series is Intel Integrated GPU, Radeon M series is AMD integrated GPU)
(I would check AMD chip later, due to its higher cost-effective performance. Beamformer is written by OpenCL for integrated GPU setting, it should work on it with little work)

For memory hardware capability:

2400MHz DDR memory is of quite low spec with high latency, compared to main stream hardware.

2400MHz DDR hardware access time delay comparison table.

	DDR Frequency (MHz)																														
CL	**2133**	**2400**	**2666**	**2800**	**3000**	**3200**	**3333**	**3400**	**3466**	**3600**	**3733**	**3866**	**4000**	**4133**	**4200**	**4266**	**4400**	**4600**	**4800**	**5000**	**5133**	**5200**	**5600**	**6000**	**6400**	**6600**	**6800**	**7200**	**7600**	**7800**	**8000**
7	6.56	5.83	5.25	5.00	4.67	4.38	4.20	4.12	4.04	3.89	3.75	3.62	3.50	3.39	3.33	3.28	3.18	3.04	2.92	2.80	2.73	2.69	2.50	2.33	2.19	2.12	2.06	1.94	1.84	1.79	1.75
8	7.50	6.67	6.00	5.71	5.33	5.00	4.80	4.71	4.62	4.44	4.29	4.14	4.00	3.87	3.81	3.75	3.64	3.48	3.33	3.20	3.12	3.08	2.86	2.67	2.50	2.42	2.35	2.22	2.11	2.05	2.00
9	8.44	7.50	6.75	6.43	6.00	5.63	5.40	5.29	5.19	5.00	4.82	4.66	4.50	4.36	4.29	4.22	4.09	3.91	3.75	3.60	3.51	3.46	3.21	3.00	2.81	2.73	2.65	2.50	2.37	2.31	2.25
10	9.38	8.33	7.50	7.14	6.67	6.25	6.00	5.88	5.77	5.56	5.36	5.17	5.00	4.84	4.76	4.69	4.55	4.35	4.17	4.00	3.90	3.85	3.57	3.33	3.13	3.03	2.94	2.78	2.63	2.56	2.50
11	10.31	9.17	8.25	7.86	7.33	6.88	6.60	6.47	6.35	6.11	5.89	5.69	5.50	5.32	5.24	5.16	5.00	4.78	4.58	4.40	4.29	4.23	3.93	3.67	3.44	3.33	3.24	3.06	2.89	2.82	2.75
12	11.25	10.00	9.00	8.57	8.00	7.50	7.20	7.06	6.92	6.67	6.43	6.21	6.00	5.81	5.71	5.63	5.45	5.22	5.00	4.80	4.68	4.62	4.29	4.00	3.75	3.64	3.53	3.33	3.16	3.08	3.00
13	12.19	10.83	9.75	9.29	8.67	8.13	7.80	7.65	7.50	7.22	6.96	6.73	6.50	6.29	6.19	6.09	5.91	5.65	5.42	5.20	5.07	5.00	4.64	4.33	4.06	3.94	3.82	3.61	3.42	3.33	3.25
14	13.13	11.67	10.50	10.00	9.33	**8.75**	**8.40**	**8.24**	8.08	7.78	7.50	7.24	7.00	6.77	6.67	6.56	6.36	6.09	5.83	5.60	5.45	5.38	5.00	4.67	4.38	4.24	4.12	3.89	3.68	3.59	3.50
15	14.06	12.50	11.25	10.71	10.00	9.38	9.00	8.82	**8.66**	**8.33**	8.04	7.76	7.50	7.26	7.14	7.03	6.82	6.52	6.25	6.00	5.84	5.77	5.36	5.00	4.69	4.55	4.41	4.17	3.95	3.85	3.75
16	15.00	13.33	12.00	11.43	10.67	10.00	9.60	9.41	9.23	8.89	**8.57**	**8.28**	8.00	7.74	7.62	7.50	7.27	6.96	6.67	6.40	6.23	6.15	5.71	5.33	5.00	4.85	4.71	4.44	4.21	4.10	4.00
17	15.94	14.17	12.75	12.14	11.33	10.63	10.20	10.00	9.81	9.44	9.11	8.79	**8.50**	**8.23**	8.10	7.97	7.73	7.39	7.08	6.80	6.62	6.54	6.07	5.67	5.31	5.15	5.00	4.72	4.47	4.36	4.25
18	16.88	15.00	13.50	12.86	12.00	11.25	10.80	10.59	10.39	10.00	9.64	9.31	9.00	**8.71**	**8.57**	**8.44**	8.18	7.83	7.50	7.20	7.01	6.92	6.43	6.00	5.63	5.45	5.29	5.00	4.74	4.62	4.50
19	17.82	15.83	14.25	13.57	12.67	11.88	11.40	11.18	10.96	10.56	10.18	9.83	9.50	9.19	9.05	8.91	8.64	8.26	7.92	7.60	7.40	7.31	6.79	6.33	5.94	5.76	5.59	5.28	5.00	4.87	4.75
20	18.75	16.67	15.00	14.29	13.33	12.50	12.00	11.76	11.54	11.11	10.72	10.35	10.00	9.68	9.52	9.38	9.09	**8.70**	8.33	8.00	7.79	7.69	7.14	6.67	6.25	6.06	5.88	5.56	5.26	5.13	5.00
21	19.69	17.50	15.75	15.00	14.00	13.13	12.60	12.35	12.12	11.67	11.25	10.86	10.50	10.16	10.00	9.85	9.55	9.13	**8.75**	**8.40**	8.18	8.08	7.50	7.00	6.56	6.36	6.18	5.83	5.53	5.38	5.25
22	20.63	18.33	16.50	15.71	14.67	13.75	13.20	12.94	12.69	12.22	11.79	11.38	11.00	10.65	10.48	10.31	10.00	9.57	9.17	8.80	**8.57**	**8.46**	7.86	7.33	6.88	6.67	6.47	6.11	5.79	5.64	5.50
23	21.57	19.17	17.25	16.43	15.33	14.38	13.80	13.53	13.27	12.78	12.32	11.90	11.50	11.13	10.95	10.78	10.45	10.00	9.58	9.20	8.96	8.85	8.21	7.67	7.19	6.97	6.76	6.39	6.05	5.90	5.75
24	22.50	20.00	18.00	17.14	16.00	15.00	14.40	14.12	13.85	13.33	12.86	12.42	12.00	11.61	11.43	11.25	10.91	10.43	10.00	9.60	9.35	9.23	**8.57**	**8.00**	7.50	7.27	7.06	6.67	6.32	6.15	6.00
28	26.25	23.33	21.01	20.00	18.67	17.50	16.80	16.47	16.16	15.56	15.00	14.49	14.00	13.55	13.33	13.13	12.73	12.17	11.67	11.20	10.91	10.77	10.00	9.33	**8.75**	**8.48**	**8.24**	7.78	7.37	7.18	7.00
30	28.13	25.00	22.51	21.43	20.00	18.75	18.00	17.65	17.31	16.67	16.07	15.52	15.00	14.52	14.29	14.06	13.64	13.04	12.50	12.00	11.69	11.54	10.71	10.00	9.38	9.09	8.82	**8.33**	7.89	7.69	7.50
32	30.00	26.67	24.01	22.86	21.33	20.00	19.20	18.82	18.47	17.78	17.14	16.55	16.00	15.49	15.24	15.00	14.55	13.91	13.33	12.80	12.47	12.31	11.43	10.67	10.00	9.70	9.41	8.89	**8.42**	8.21	8.00
34	31.88	28.33	25.51	24.29	22.67	21.25	20.40	20.00	19.62	18.89	18.22	17.59	17.00	16.45	16.19	15.94	15.45	14.78	14.17	13.60	13.25	13.08	12.14	11.33	10.63	10.30	10.00	9.44	8.95	**8.72**	**8.50**
36	33.76	30.00	27.01	25.71	24.00	22.50	21.60	21.18	20.77	20.00	19.29	18.62	18.00	17.42	17.14	16.88	16.36	15.65	15.00	14.40	14.03	13.85	12.86	12.00	11.25	10.91	10.59	10.00	9.47	9.23	9.00
38	35.63	31.67	28.51	27.14	25.33	23.75	22.80	22.35	21.93	21.11	20.36	19.66	19.00	18.39	18.10	17.82	17.27	16.52	15.83	15.20	14.81	14.62	13.57	12.67	11.88	11.52	11.18	10.56	10.00	9.74	9.50
40	37.51	33.33	30.01	28.57	26.67	25.00	24.00	23.53	23.08	22.22	21.43	20.69	20.00	19.36	19.05	18.75	18.18	17.39	16.67	16.00	15.59	15.38	14.29	13.33	12.50	12.12	11.76	11.11	10.53	10.26	10.00
RATE:		Memory clock frequency in MHz																													
CL:		CAS Latency - It tells us how many clock cycles the memory will delay to return requested data.																													

The EchoSight-Pro performance is purely powered by system theorical and engineering architecture design, instead of adding hardware resources. It is the **technique moat for cost effective** ultrasound system.

7. Author foreshows

Version 0.5.0 is a critical important update. This version makes it the fastest integrated GPU ultrasound software framework by now.
Version 0.5.1 provides research interface for academic usage; it will be very handy for guys. Also, algorithm update is made to minimize integrated GPU beamformer artifacts.

If somebody surpassed the record later, I would be happy to see that.
As I am not doing it in traditional R&D way; only one person, not full time, I do believe somebody is possible to achieve better if they have a solid mature system engineering team in commercial company.

Next step, I might fix some existing bugs, expand to other probes, such as curve or phase, may be try to improve efficiency further more? I am not sure neither. . .